

Next Generation Data Centers and Energy Management Systems

Lecture 2C: Heat Transfer

Prof. Thomas M. Schutzius

Department of Mechanical Engineering

Fall 2026

Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and **Heat Transfer**
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Introduce Project 1
- Define heat exchanger effectiveness

Typical layout of a data center facility

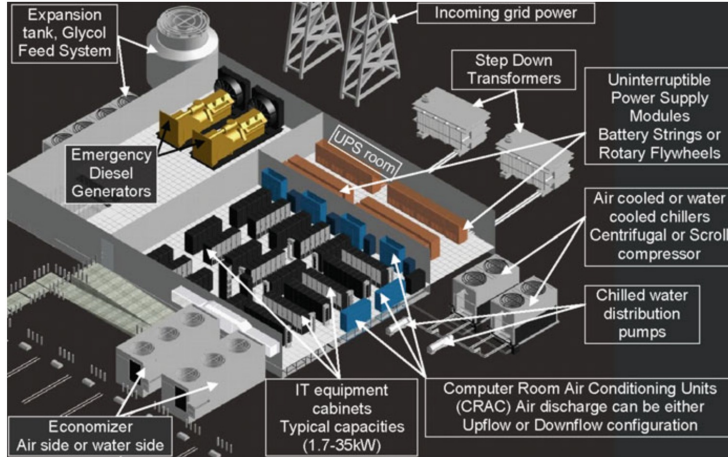


Figure 1: (Figure 1.1 in Ref. [2]).

Seminar 1

- Dr. Imran Latif. Johnson Controls International (JCI)
- Title: From Chip to Chiller: LLM Agents as the Next Efficiency Lever in AI Data Centers
- Date: Wednesday, 9/16/26
- Time: 9:00 am (Berkeley Time)
- Location: Etcheverry Hall 3110
- It will be recorded and posted to bCourses. Please do not distribute.

Project 1

Project 1: Properties

Heat Exchanger Effectiveness

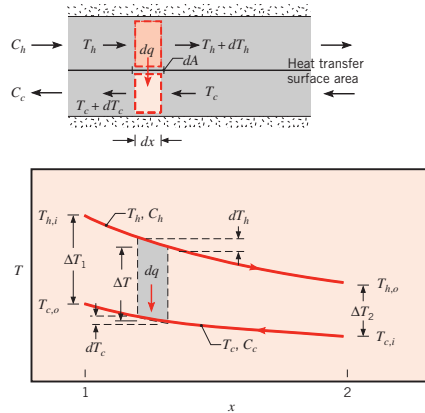


Figure 2: Temperature distributions for a counterflow heat exchanger (Figure 11.8 in Ref. [1]).

Heat Exchanger Effectiveness

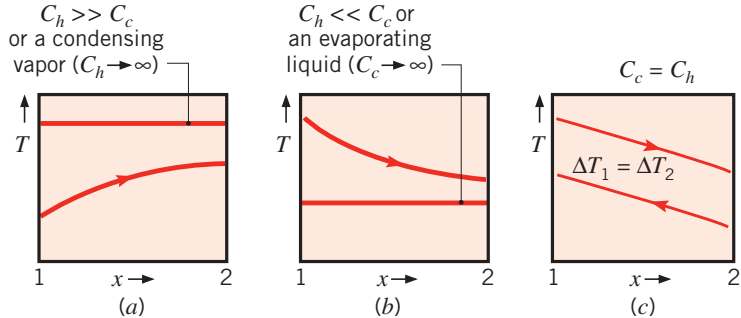
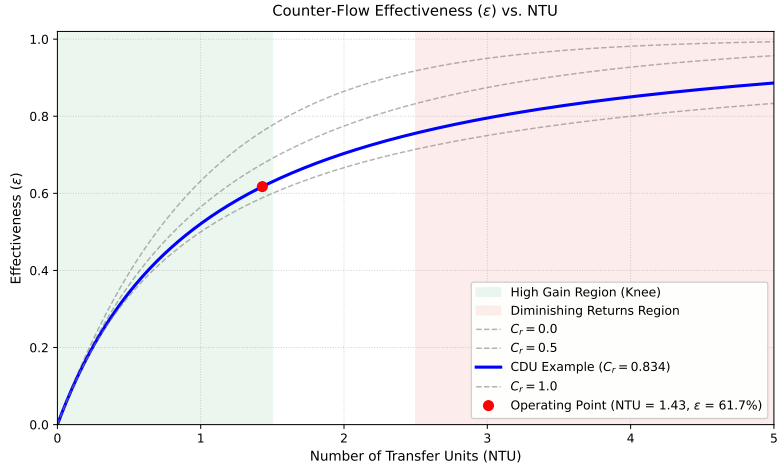


Figure 3: (a) $C_h \gg C_c$ and (b) $C_h \ll C_c$, (c) A counterflow heat exchanger with equivalent fluid heat capacities ($C_h = C_c$) (Figure 11.9 in Ref. [1]).

Counter-flow Heat Exchanger Effectiveness



References I

- [1] Frank P Incropera, David P DeWitt, Theodore L Bergman, and Adrienne S Lavine.
Introduction to heat transfer.
John Wiley & Sons, 2007.
- [2] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.